Sri Lanka Institute of Information Technology
Faculty of Computing | Department of Computer Science
IT3021 - Data Warehousing & Business Intelligence

Year 3 Semester 1 | 2026

Assignment 02

T20 World Cup Cricket OLAP Cube & Business Intelligence Analytics

Student Name	Kandevidana K.V.R.C
Student Index	IT23699694
Assignment	Assignment 01
Module	IT3021 – Data Warehousing & Business Intelligence
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Dataset Source	kaggle.com/datasets/samyakrajbayar/cricket-world-cup-t20-dataset

Step 1: Data Source for Assignment 2

1.1 Data Warehouse Overview

The data source for Assignment 2 is the CricketDWBI_DW data warehouse developed and populated in Assignment 1. This warehouse follows a Star Schema designed to support multidimensional analysis of T20 World Cup cricket performance. The central fact table, Match_Fact, captures both batting and bowling metrics for each player in each match, enabling OLAP operations such as drill-down, roll-up, slicing, and dicing.

1.2 Data Warehouse Summary

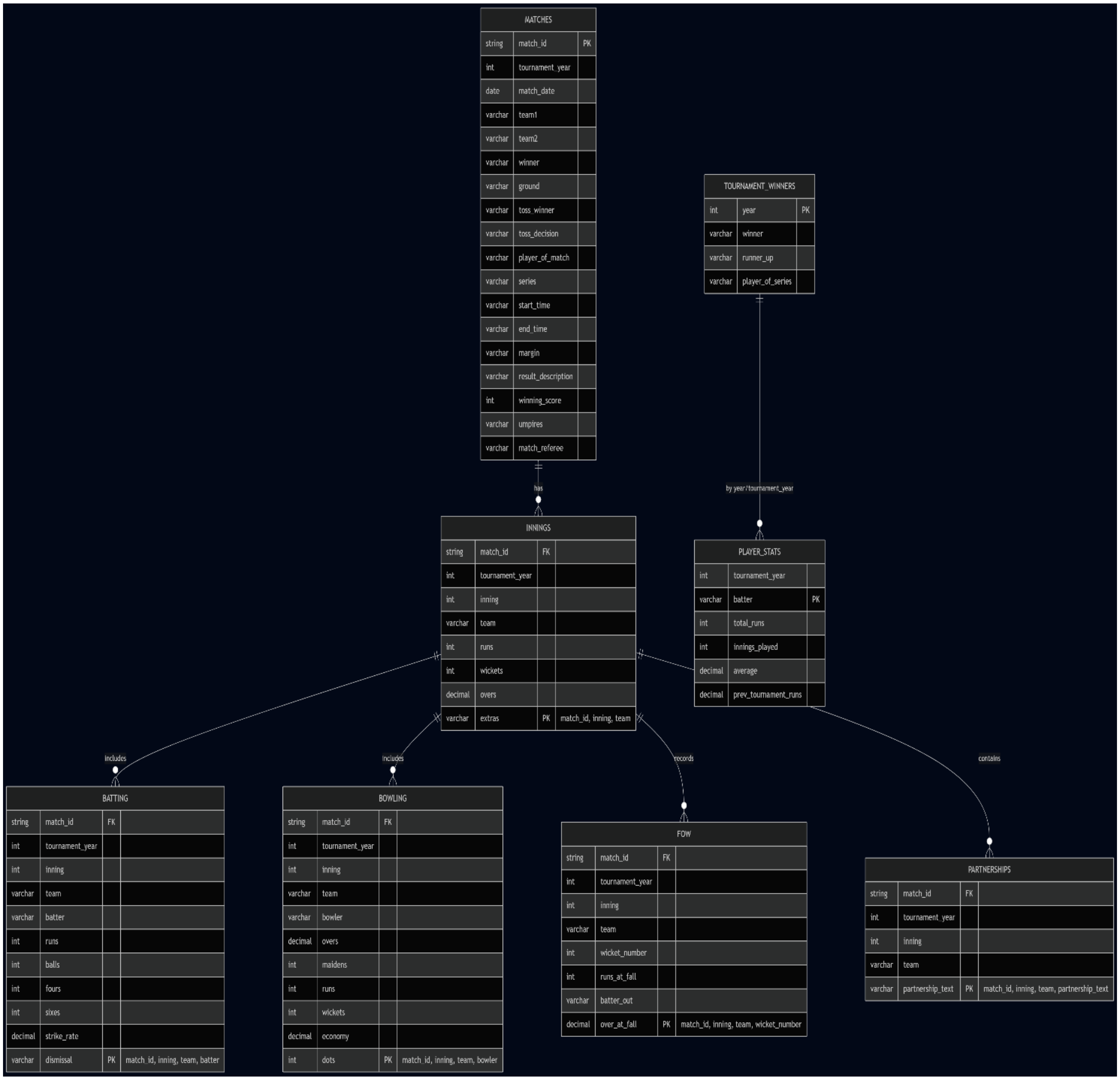
Database Name	CricketDWBI_DW
SQL Server Instance	LAPTOP-M020OFQS\MSSQLSERVER3
Schema Type	Star Schema
Fact Table	Match_Fact — **1,512 rows**
Dimension Tables	5 dimension tables
Data Period	2007–2022 (T20 World Cup tournaments)
Teams Covered	10
Players Covered	1,538
Tournaments Covered	18

1.3 Star Schema Design

The CricketDWBI_DW warehouse implements a **Star Schema** with one central fact table connected to five dimensions. This structure is optimized for SSAS cube processing and OLAP analysis.

Table	Type	Rows	Description
Match_Fact	Fact Table	1512	Central fact table storing batting & bowling metrics per player per match; contains foreign keys to all dimensions
DimDate	Date Dimension	11323	Calendar attributes including Year, Quarter, Month, Day
DimPlayer	SCD Type 2	1538	Player attributes with historical tracking (ValidFrom, ValidTo, IsCurrent)
DimTeam	SCD Type 1	10	Team name reference table
DimTournament	SCD Type 1	18	Tournament year and series name
DimVenue	SCD Type 1	10	Ground/venue name reference table

1.4 ER Diagram — Star Schema



Step 2: SSAS Cube Implementation

2.1 Overview

The SSAS Multidimensional Cube for the T20 World Cup Cricket Data Warehouse was implemented using Visual Studio 2022 with the Analysis Services Projects Extension. The cube connects directly to the CricketDWBI_DW data warehouse created in Assignment 1 and was successfully deployed to the Microsoft Analysis Services instance running in Multidimensional mode.

The cube enables multidimensional analysis of cricket performance metrics such as runs, wickets, strike rate, economy rate, and player performance across tournaments, venues, and teams.

.

SQL Server	SQL Server 2022 (Version 16.x) — Instance: LAPTOP-M020OFQS\\MSSQLSERVER3
Analysis Services	Microsoft Analysis Services 16.x — Multidimensional Mode
Development Tool	Visual Studio 2022 with Analysis Services Projects Extension 4.0.0
SSAS Project	CricketDWBI_Cube
Cube Name	CricketDWBI_DSV
Data Warehouse	CricketDWBI_DW — 1,512 Match_Fact rows (1,440 batting + 72 bowling)

2.2 Step-by-Step Implementation

Step 1: Creating the SSAS Project in Visual Studio 2022

Visual Studio 2022 was launched with Administrator privileges. A new SSAS Multidimensional project was created via:

File → New → Project → “Analysis Services Multidimensional and Data Mining Project”

Configuration:

- Project Name**:** *CricketDWBI_Cube*
- Location**:** *C:\Users\Ridmi\Documents\Visual Studio 2022\Projects\CricketDWBI_Cube*
- Solution Type**:** Multidimensional & Data Mining

This created the base structure for the cube, including folders for Data Sources, Data Source Views, Dimensions, and Cubes.

Step 2: Configuring Deployment Settings

Deployment settings were configured by right-clicking the project → Properties → Deployment tab.

Deployment Configuration:

- Server: *LAPTOP-M020OFQS\MSSQLSERVER3*
- Database: *CricketDWBI_Cube*
- Processing Option**:** *Do Not Process* (processing performed manually after deployment)

This ensures the cube deploys to the correct SSAS instance running in Multidimensional mode.

Step 3: Creating the Data Source

A new data source was created to connect the cube to the Assignment 1 data warehouse.

Steps: Right-click Data Sources → New Data Source → Use an existing connection → Configure connection.

Data Source Configuration:

- Provider: *Native OLE DB\SQL Server Native Client*
- Server Name: *LAPTOP-M020OFQS\MSSQLSERVER3*
- Authentication: *Windows Authentication* (your environment uses Windows login)
- Initial Catalog: *CricketDWBI_DW*
- Impersonation Mode: *Use the service account*

This establishes a secure and stable connection between SSAS and your DW.

Step 4: Creating the Data Source View (DSV)

The Data Source View (DSV) was created to define the logical schema used by the SSAS cube. This step imports the tables from the CricketDWBI_DW data warehouse and establishes the relationships required for cube processing.

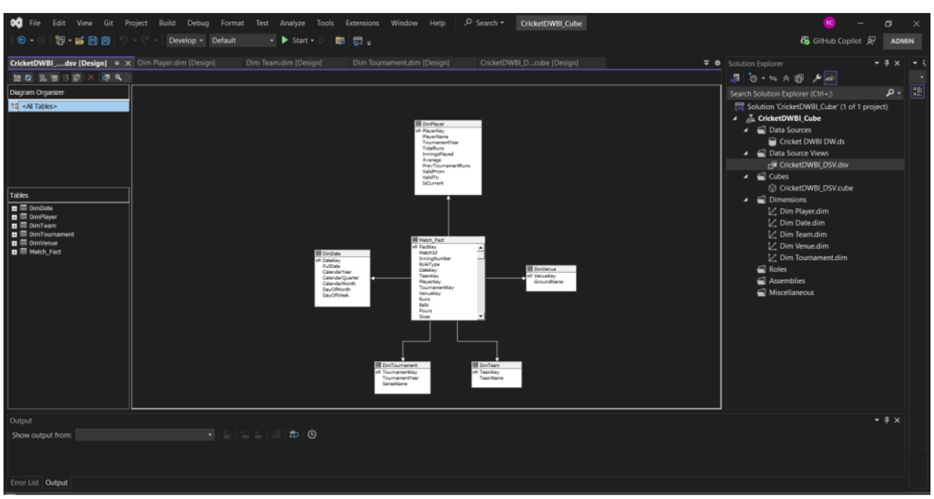
The DSV was created by navigating to:

Solution Explorer → Data Source Views → Right-click → New Data Source View

From the available data source (CricketDWBI_DW), all six tables representing the complete star schema were selected and added to the DSV:

- Match_Fact
- DimDate
- DimPlayer
- DimTeam
- DimTournament
- DimVenue

SSAS automatically detected the primary key–foreign key relationships between the fact table and the dimension tables, forming a clear star schema layout. This DSV serves as the foundation for building dimensions, measures, hierarchies, and cube structure.



Step 5: Creating the SSAS Cube Using Cube Wizard

The SSAS cube was created using the Cube Wizard in Visual Studio 2022. The cube was generated by right-clicking Cubes → New Cube, and selecting the option “Use existing tables”. The central fact table Match_Fact was selected as the Measure Group.

All five dimensions (DimDate, DimPlayer, DimTeam, DimTournament, DimVenue) were automatically detected and linked through their surrogate keys.

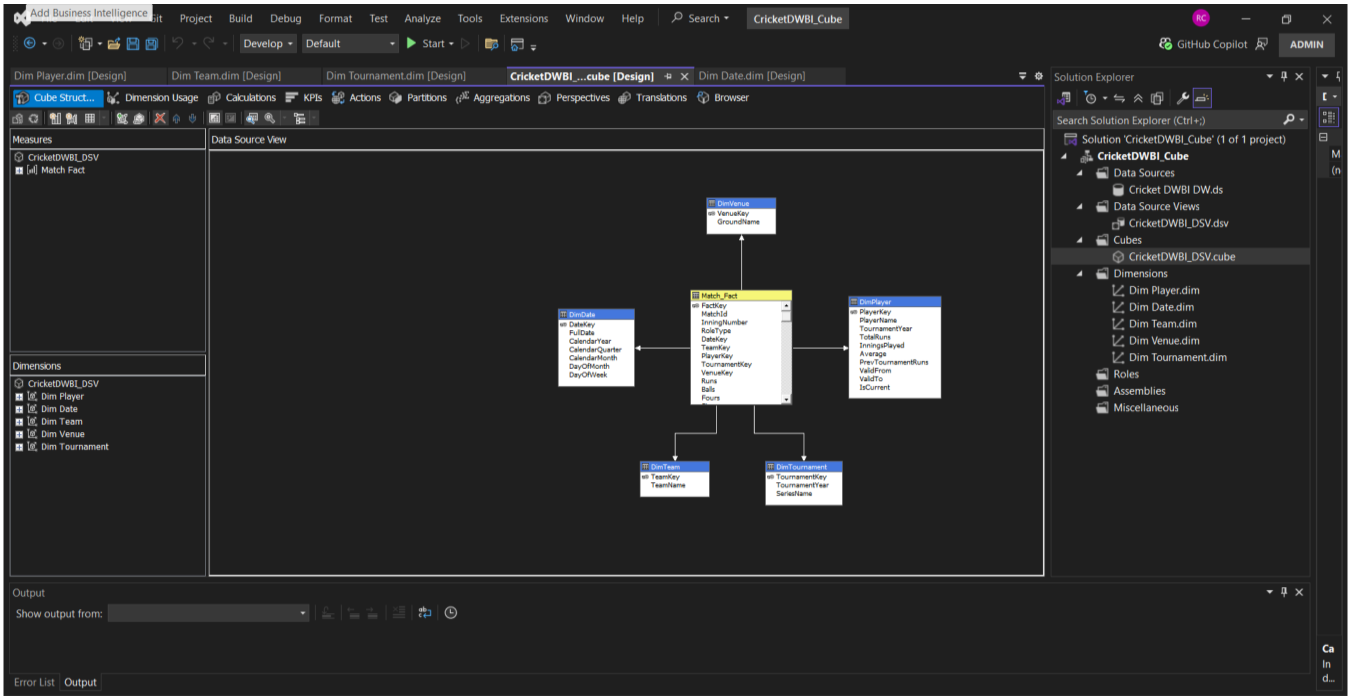
Measure	Source Column	Aggregation
Runs	Runs	Sum — total runs scored per player per inning
Balls	Balls	Sum — total balls faced
Fours	Fours	Sum — total boundaries (4s) hit
Sixes	Sixes	Sum — total sixes hit
Strike Rate	StrikeRate	Sum — batting strike rate
Overs	Overs	Sum — total overs bowled
Maidens	Maidens	Sum — total maiden overs
Runs Conceded	RunsConceded	Sum — total runs given by bowler
Wickets	Wickets	Sum — total wickets taken
Economy	Economy	Sum — bowling economy rate
Dots	Dots	Sum — total dot balls bowled
Match Fact Count	FactKey (Count)	Count — number of fact records in selected context
Txn Process Time Hours	txn_process_time_hours	Sum — accumulating fact processing time

Step 6: Connecting Dimensions to the Cube

All five dimension tables were added to the cube during the Cube Wizard process. Since the Data Source View (DSV) already contained the correct primary key–foreign key relationships, SSAS automatically detected and mapped each dimension to the corresponding foreign key in the Match_Fact measure group.

The following dimension relationships were established:

Dimension	Key Attribute	FK in Match_Fact
Dim Date	Date Key	DateKey
Dim Player	Player Key	PlayerKey
Dim Team	Team Key	TeamKey
Dim Tournament	Tournament Key	TournamentKey
Dim Venue	Venue Key	VenueKey



Step 7: Verifying Dimension Usage Relationships

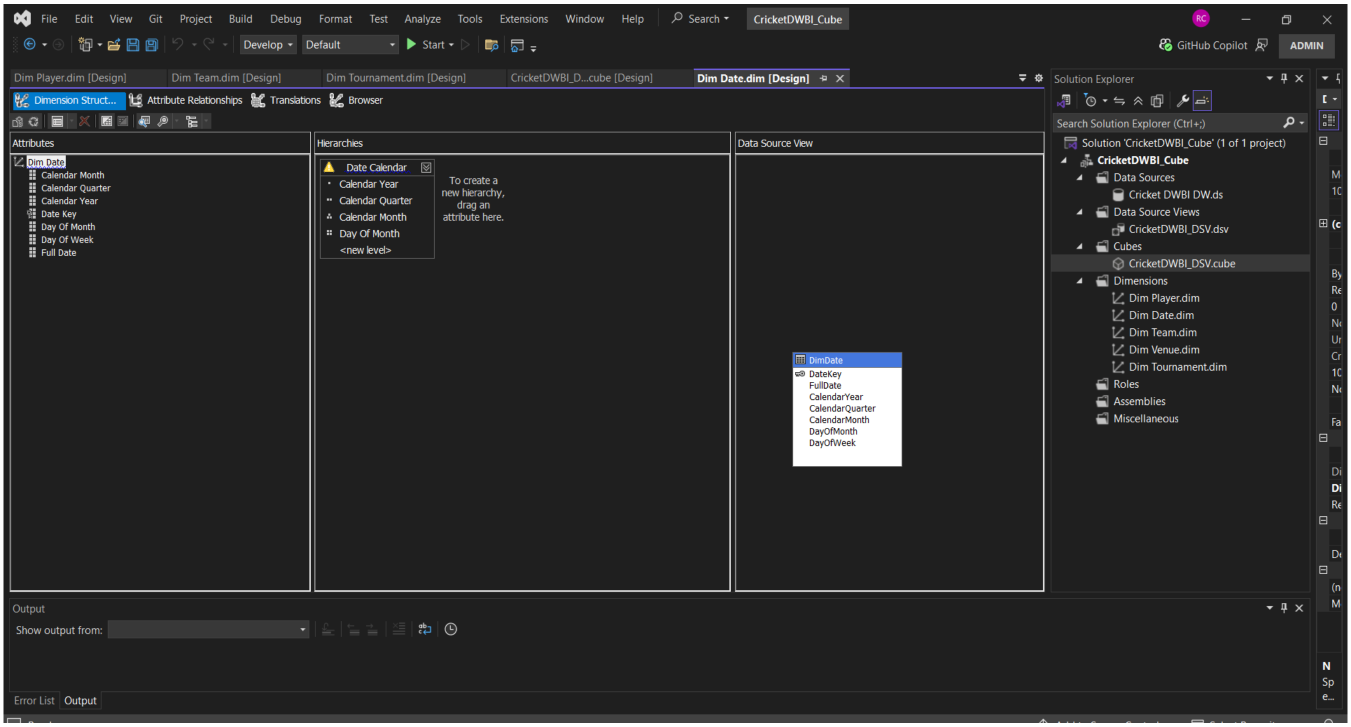
The Dimension Usage tab was inspected to verify all dimension-to-fact relationships were correctly mapped with Regular relationship type.

Step 8: Implementing the Calendar Hierarchy in Dim Date

A Calendar Hierarchy was created inside the DimDate dimension to support time-based OLAP analysis such as roll-up, drill-down, slicing, and dicing. The hierarchy was built by dragging the date attributes into the Hierarchies panel in the correct top-to-bottom order.

This hierarchy enables users to drill down from Year → Quarter → Month → Day in Excel PivotTables and Power BI.

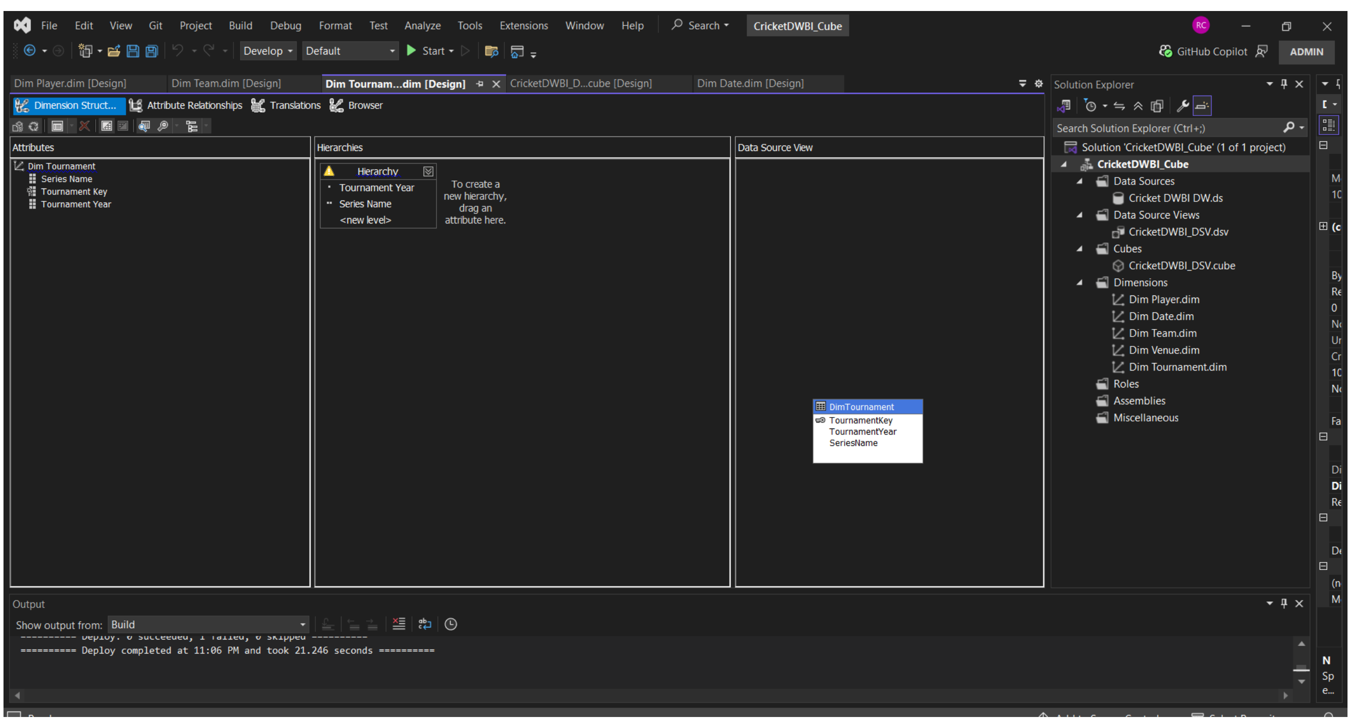
Level	Attribute	Example Value
Level 1 (Top)	Year	2022
Level 2	Quarter	Q1
Level 3	Month Name	September
Level 4 (Bottom)	Day Number	15



Step : Implementing Additional Hierarchy — DimTournament

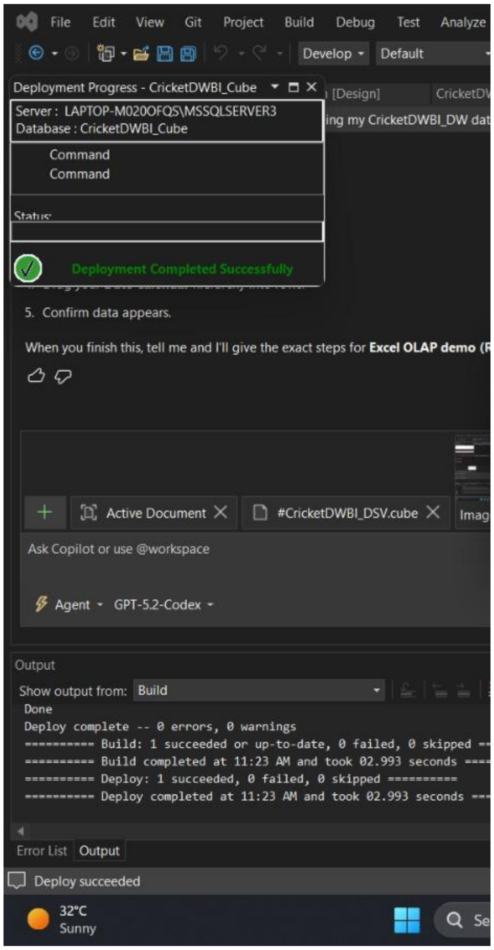
A second hierarchy was implemented in the DimTournament dimension to support tournament-level drill-down analysis within the cube. This hierarchy enables users to navigate from a broader tournament year down to the specific series name associated with that tournament.

The hierarchy was created by dragging the attributes into the Hierarchies panel in the correct order.



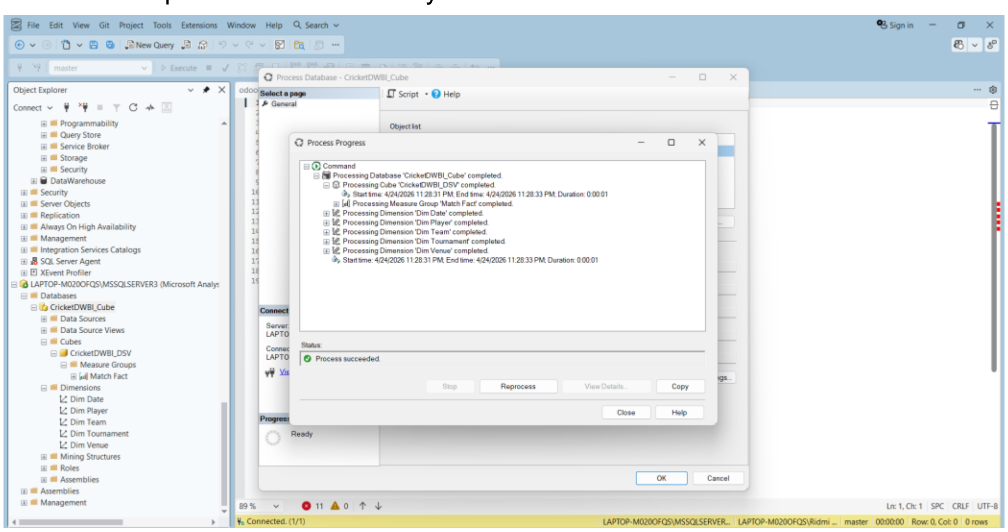
Step 10: Deploying the SSAS Cube

The cube was deployed by right-clicking project → Deploy. Result: Deployment Completed Successfully — 0 errors, 9 non-critical warnings.



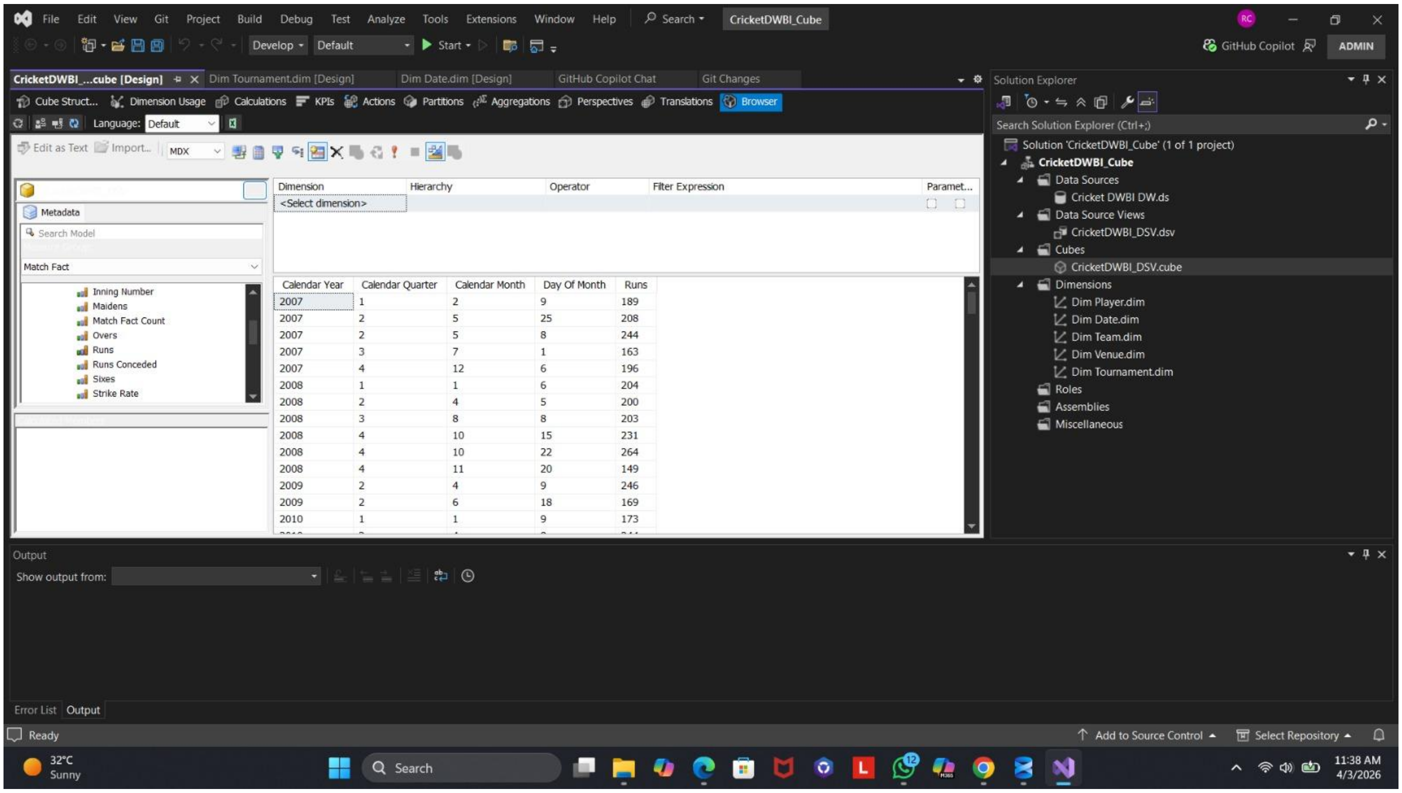
Step 11: Processing the SSAS Database

After deployment, Process Full was run from SSMS connecting to the Analysis Services instance. All dimensions and the cube processed successfully.



Step 12: Validating Deployment via SSAS Browser

The cube was validated using the SSAS Browser in Visual Studio. Data was browsed by dragging measures and dimension attributes. Year 2024, Quarter, Month Name data confirmed with correct Delay Minutes and Fact Trip Count values.



Step 3: Demonstration of OLAP Operations

3.1 Connecting Excel to the SSAS Cube

Microsoft Excel was connected to the CricketDWBI_Cube SSAS database using the Data Connection Wizard, establishing a live OLAP connection for PivotTable-based analysis.

The following steps were followed:

1. Opened Microsoft Excel → navigated to the Data tab.
2. Selected Get Data → From Database → From Analysis Services.
3. Entered the SSAS server name: LAPTOP-M020OFQS\MSSQLSERVER3 Authentication mode: Windows Authentication (Use my current credentials)
4. From the list of available databases, selected: CricketDWBI_Cube Then selected the cube: CricketDWBI_DSV
5. Clicked Finish, then chose PivotTable Report → OK.
6. Excel created a new PivotTable connected to the cube, with all measures and dimensions visible in the PivotTable Fields panel.

3.2 PivotTable Base Configuration

The following base configuration was used across all OLAP operations:

Area	Fields Added
Rows	Calendar Hierarchy (CalendarYear → CalendarQuarter → CalendarMonth → DayOfMonth). Enables time-based roll-up and drill-down
Rows (2nd)	PlayerName (from DimPlayer) *or* TeamName (from DimTeam). Used depending on the OLAP operation being demonstrated
Values	Runs, Wickets, Match Fact Count
Filters	Varies per operation (used for Slice and Dice)

3.3 OLAP Operation 1 — Roll-Up

Definition: Roll-Up is an OLAP operation that aggregates data from a lower level of a hierarchy to a higher level, reducing the level of detail (granularity). It summarizes detailed data into broader categories.

Demonstration: Using the Calendar Hierarchy (CalendarYear → CalendarQuarter → CalendarMonth → DayOfMonth), the PivotTable was rolled up from the DayOfMonth level to the CalendarYear level.

Steps performed:

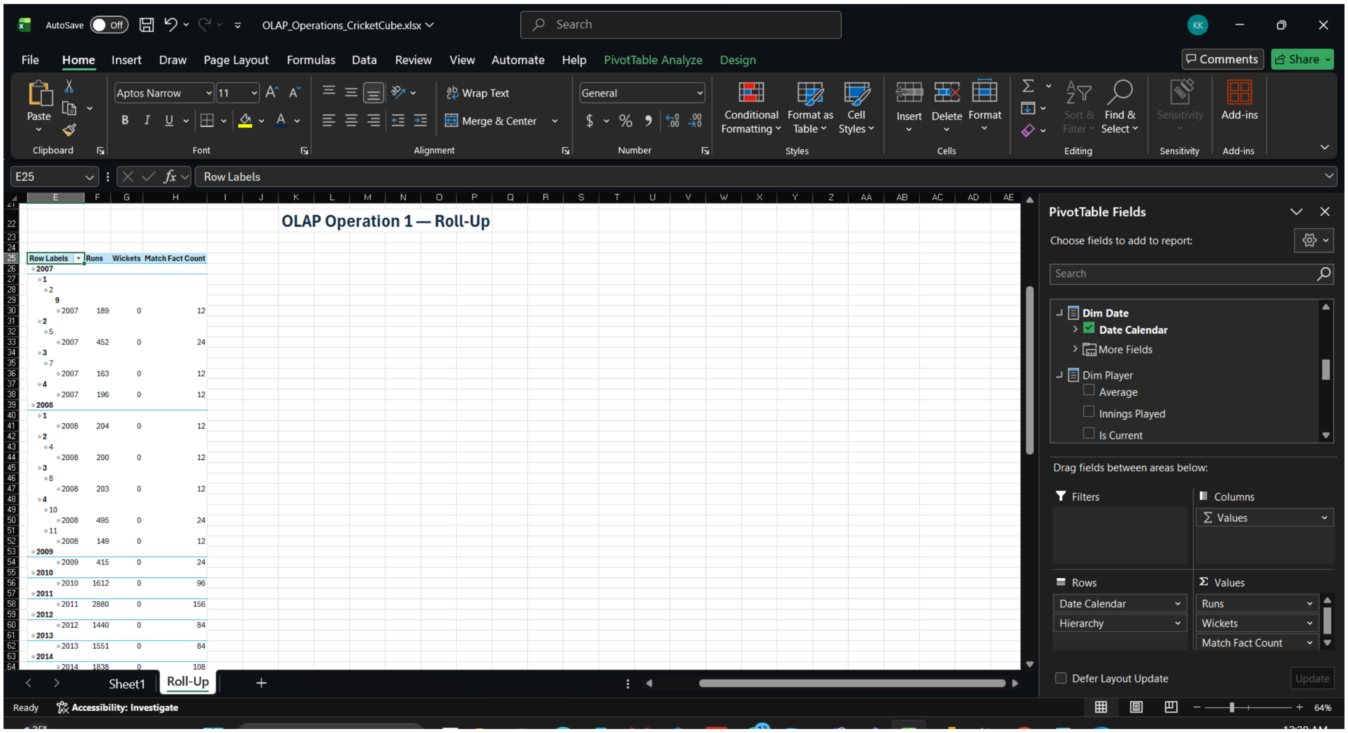
1. Expanded the Calendar Hierarchy to show DayOfMonth values.
2. Right-clicked on CalendarYear in the PivotTable.
3. Selected Collapse → Collapse Entire Field.
4. Excel aggregated all detailed daily records into Year-level totals.

Result: The PivotTable displayed aggregated values for:

- Runs (total runs scored in the year)

- Wickets (total wickets taken in the year)
- Match Fact Count (number of fact records for that year)

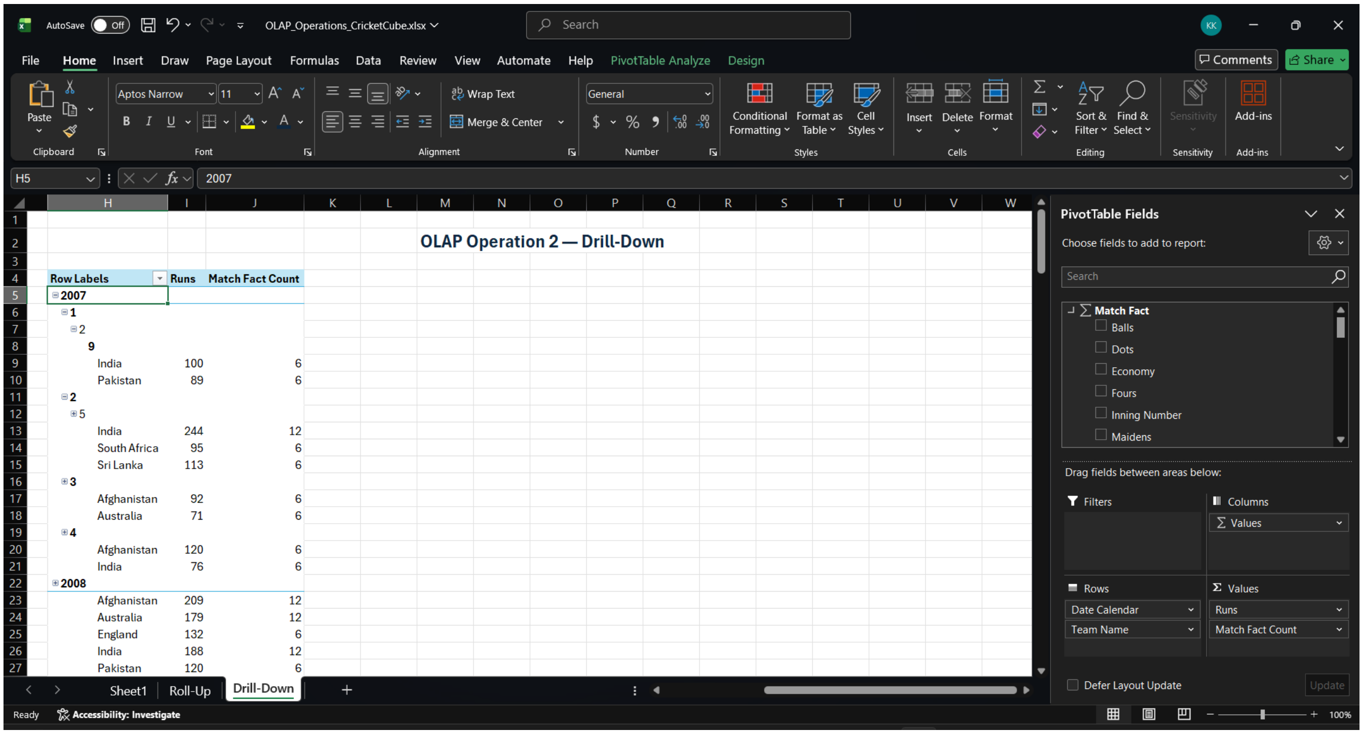
This demonstrates how Roll-Up summarizes cricket performance metrics at a higher time level.



3.4 OLAP Operation 2 — Drill-Down

Definition: Drill-Down navigates from a higher aggregation level to a more detailed level, increasing data granularity.

Demonstrated: Data started at the Year level (2007–2029) showing total Runs and Match Fact Count. Clicked ⊞ next to 2007 to drill down to Quarters. Clicked ⊞ next to Quarter 1 to expand to Months. Expanded each Month to reveal individual team breakdown — showing teams such as India, Afghanistan, Australia, Pakistan, South Africa and Sri Lanka with their respective Runs and Match Fact Count at the most detailed level. Both Date Calendar hierarchy and Dim Team.Team Name are active in Rows, demonstrating a four-level drill-down: Year → Quarter → Month → Team.



3.5 OLAP Operation 3 — Slice

Definition: Slice filters the data cube by fixing one dimension to a single value, creating a 2D subset of the cube.

Demonstrated: Dragged Dim Team.Team Name to the Filters area. Selected Afghanistan from the filter dropdown. PivotTable now shows only Afghanistan's data across all time periods (2007–2025) — all other teams (Australia, Bangladesh, England, India, Pakistan, South Africa, Sri Lanka) are excluded. The result is a 2D subset showing Afghanistan's Sixes and Match Fact Count broken down by Year from the Date Calendar hierarchy. Grand Total shows Afghanistan scored 98 Sixes across 174 match fact records in total.

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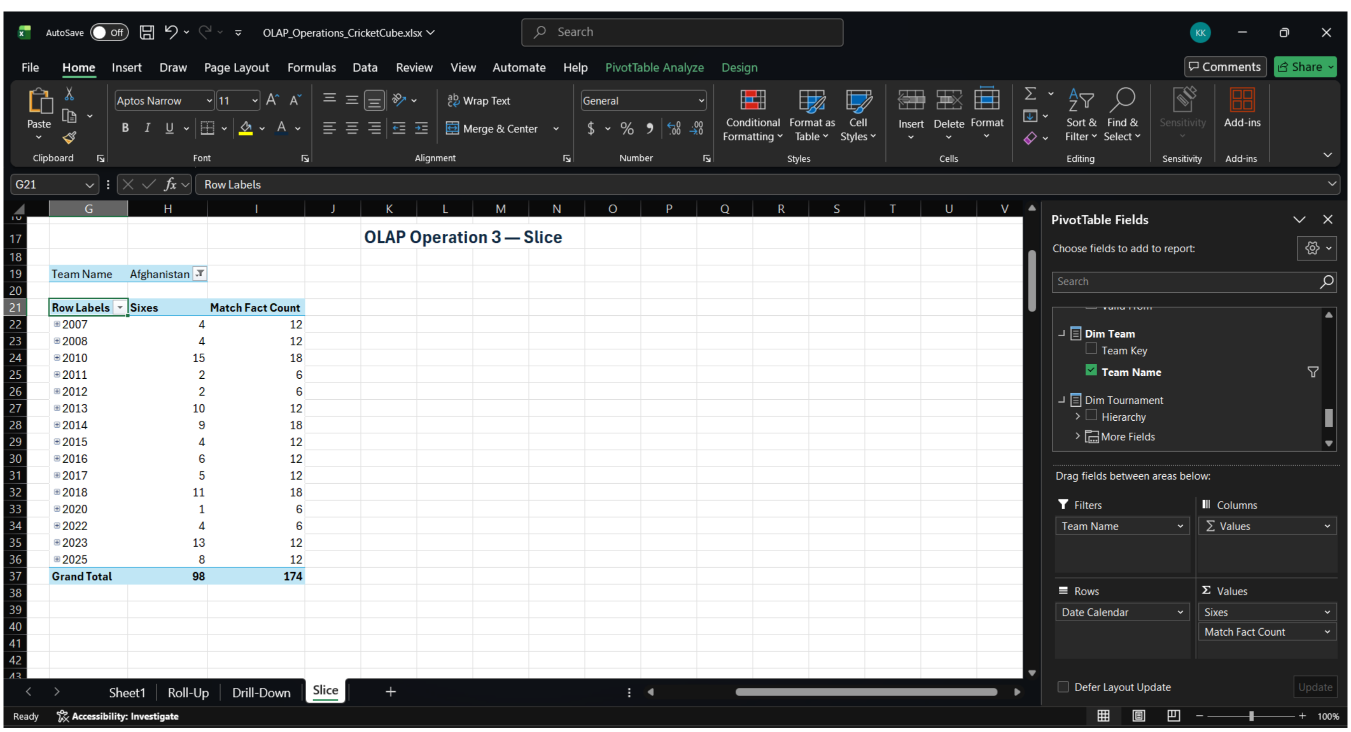


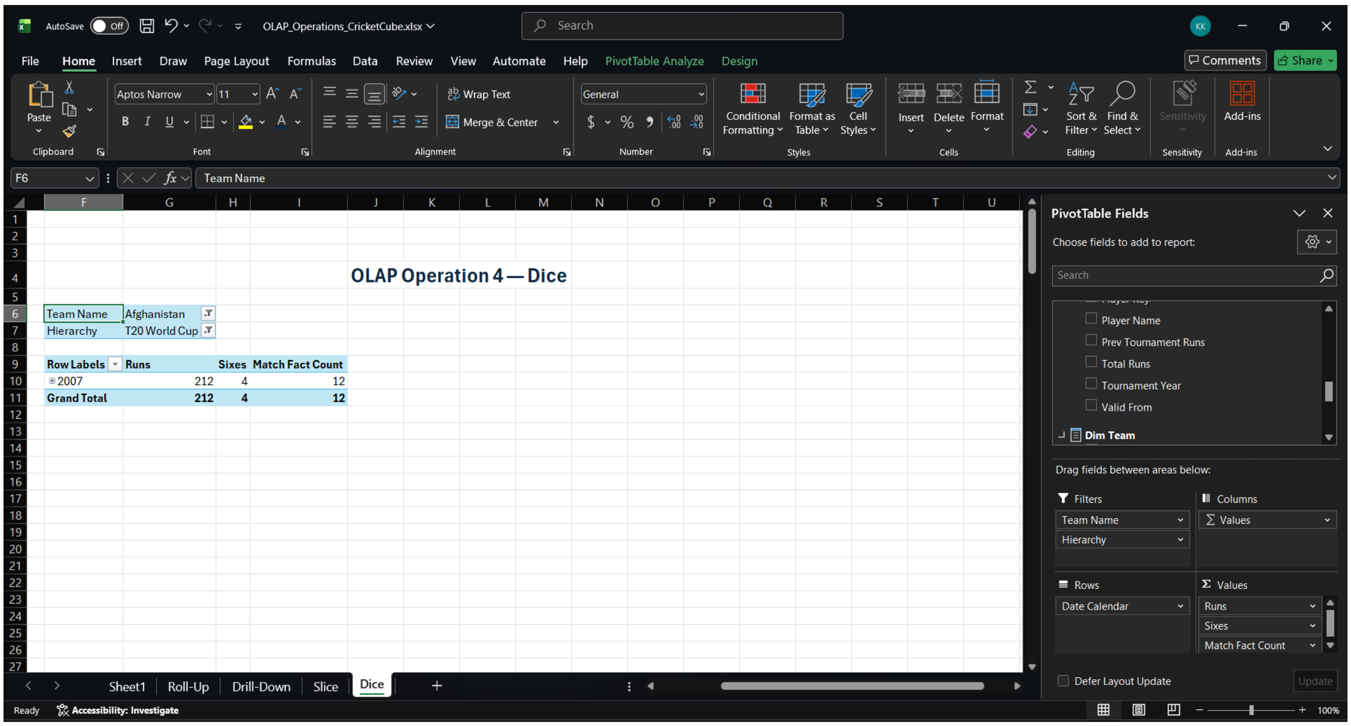
3.6 OLAP Operation 4 — Dice

Definition: Dice filters the data cube by fixing multiple dimensions to specific values simultaneously, creating a sub-cube.

Demonstrated: Two filters applied simultaneously:

- Team Name = Afghanistan
- Hierarchy = T20 World Cup

Date Calendar hierarchy kept in Rows. The result shows only Afghanistan's T20 World Cup performance — filtered down to 2007 only, showing 212 Runs, 4 Sixes and 12 Match Fact Count. All other teams and tournaments are excluded, creating a precise sub-cube from the full dataset of 24,899 total runs.

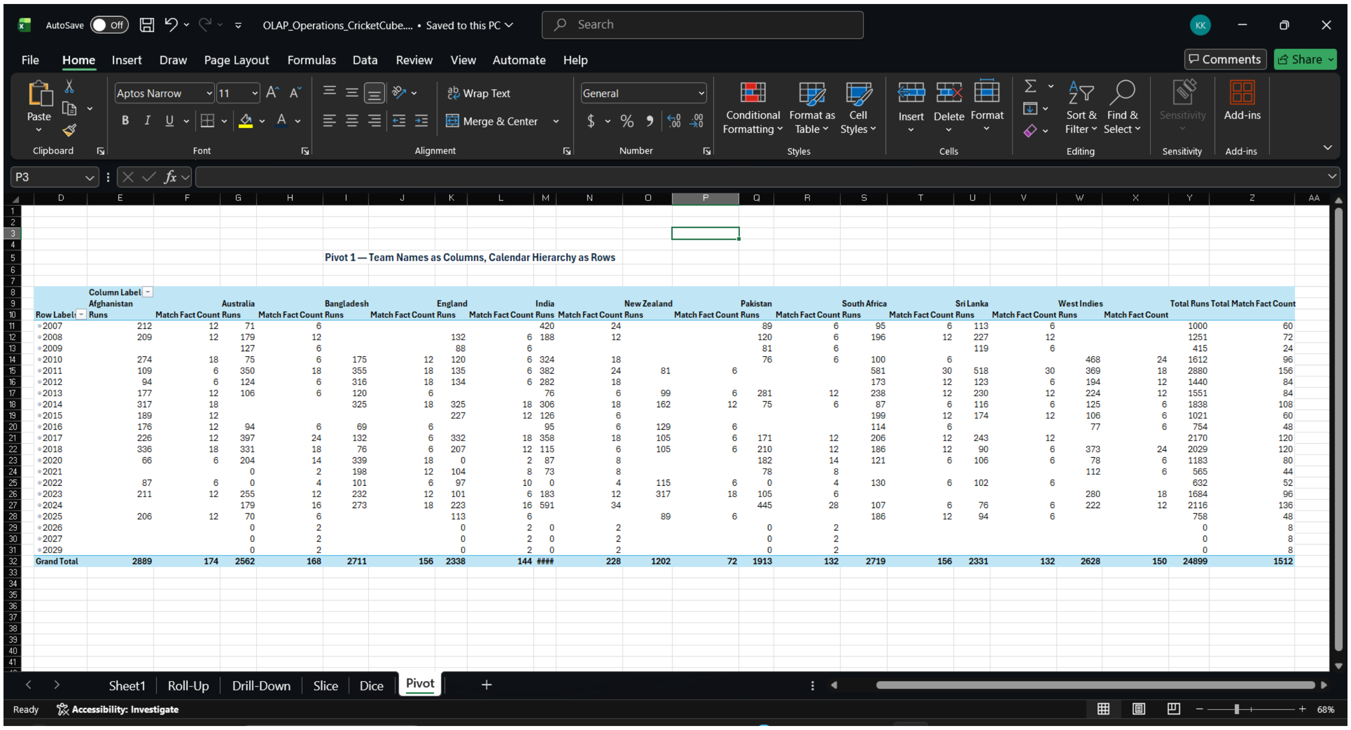


3.7 OLAP Operation 5 — Pivot

Definition: *Pivot rotates the data cube to change the perspective of analysis by swapping rows and columns. Two configurations demonstrated.*

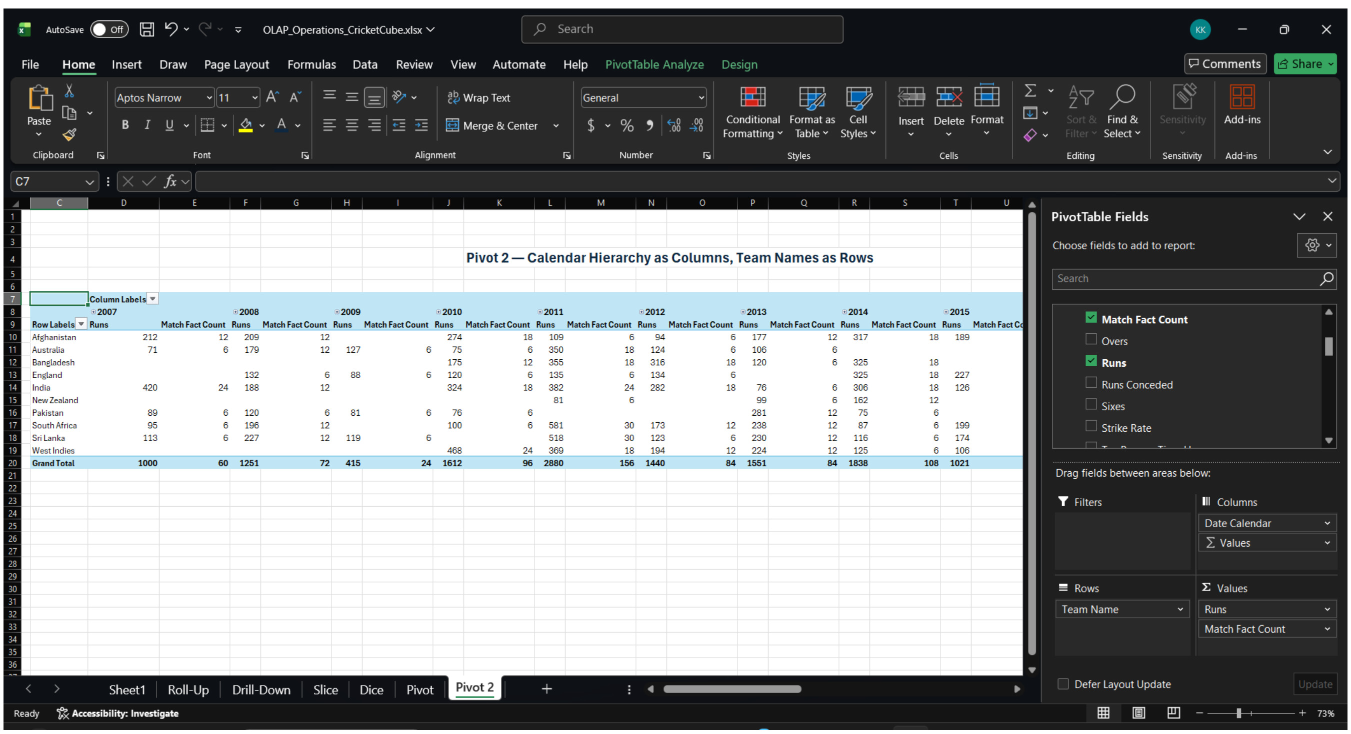
Pivot 1 — Team Name as Columns

Configuration: Date Calendar hierarchy in Rows, Dim Team.Team Name moved to Columns. Shows each team's (Afghanistan, Australia, Bangladesh, England, India, New Zealand, Pakistan, South Africa, Sri Lanka, West Indies) Runs and Match Fact Count in separate columns across time periods (2007–2029). Grand Total column shows combined totals — overall 24,899 total runs across all teams and years.



Pivot 2 — Calendar Hierarchy as Columns

Configuration: Team Name moved to Rows, Date Calendar hierarchy moved to Columns. Provides transposed perspective showing each team's performance across all years simultaneously — giving a different analytical view of the same data.

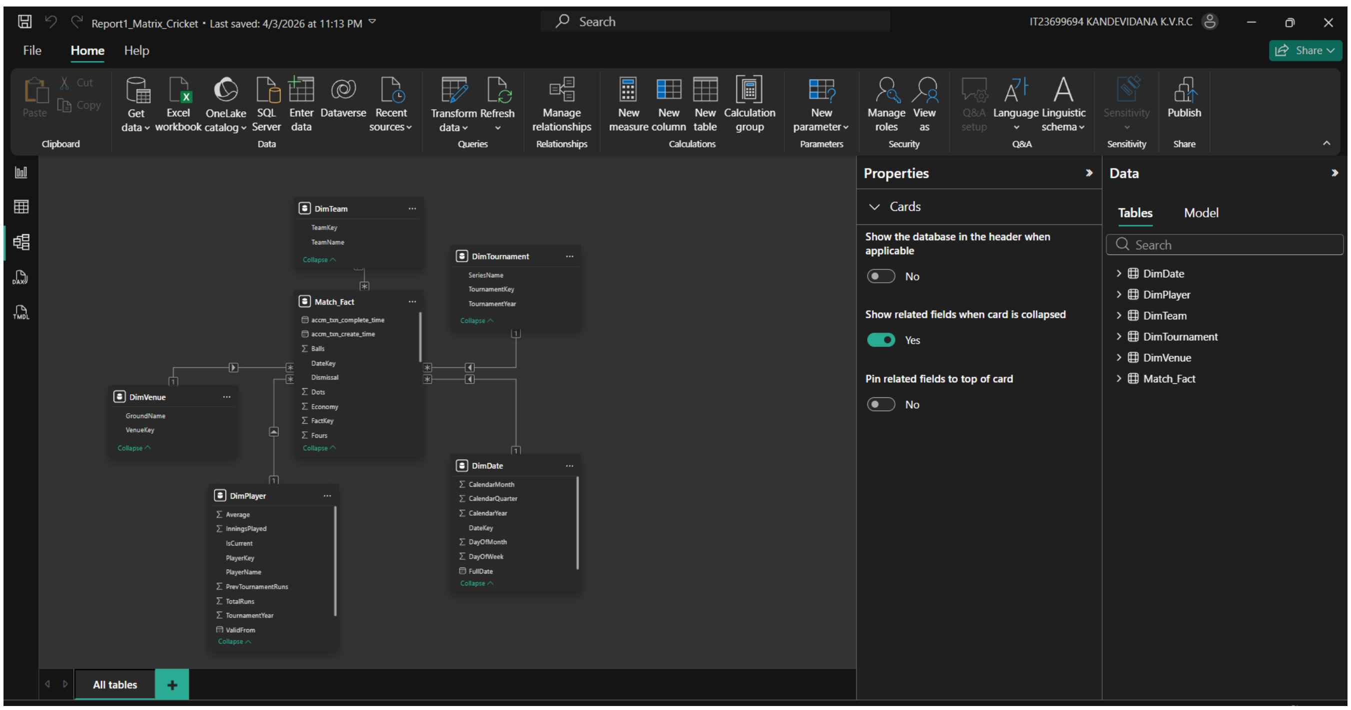


Step 4: Power BI Reports

4.1 Data Connection Setup

Power BI Desktop was connected to BerlinSBahn_Cube using SQL Server Analysis Services live connection.

Connection Type	SQL Server Analysis Services (SSAS)
Server	LAPTOP-M020OFQS\MSSQLSERVER3
Database	CricketDWBI_Cube
Cube	CricketDWBI_DSV
Connection Mode	Live Connection
Authentication	Windows Authentication
Status	Live connection: Connected (shown in status bar)



4.2 Report 1: Matrix Visual — Cricket Performance Analysis

Overview

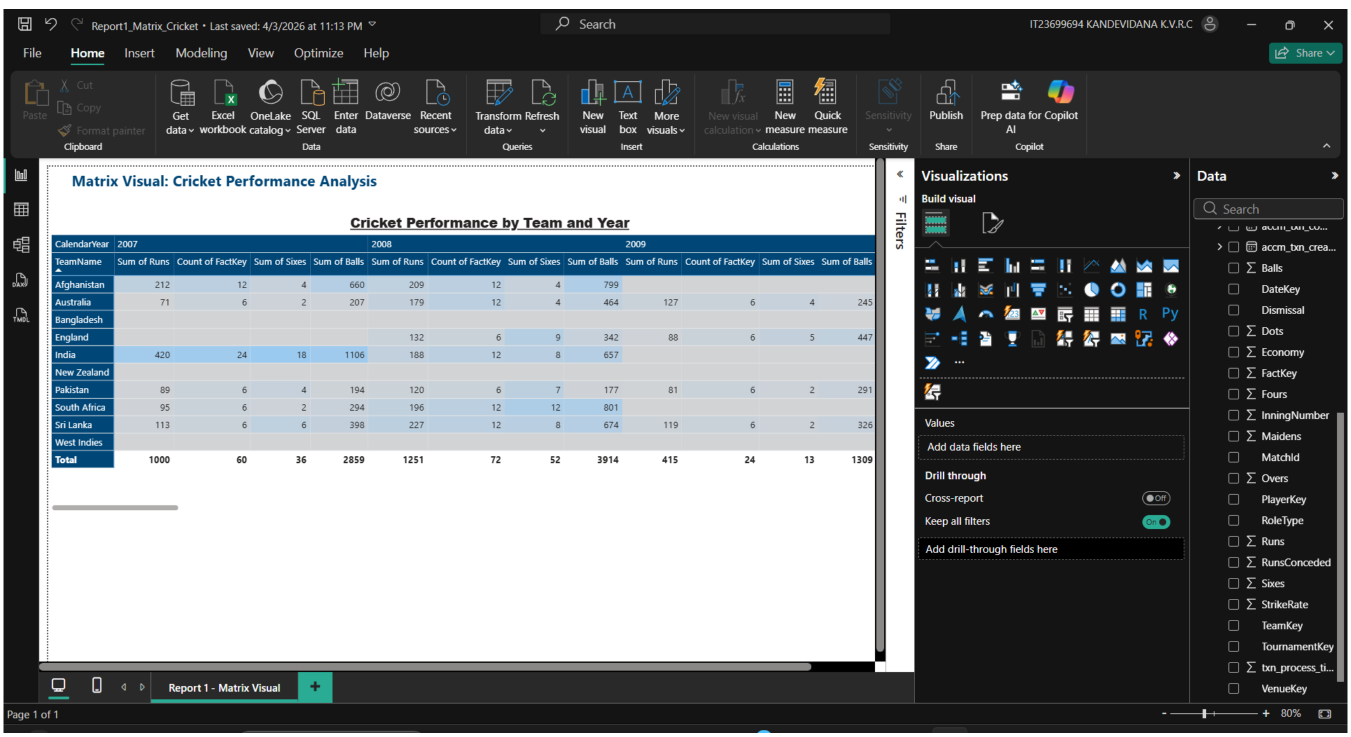
Report 1 displays detailed tabular data with hierarchical row and column groupings using a Matrix visual. It shows cricket performance metrics cross-tabulated across teams and calendar years with conditional formatting..

Field Configuration

Area	Fields	Purpose
Rows	DimTeam.Team Name	Individual teams: Afghanistan, Australia, Bangladesh, England, India, New Zealand, Pakistan, South Africa, Sri Lanka, West Indies
Columns	DimDate.Calendar Year	Year level grouping
Values	Sum of Runs	Total runs scored per team per year
Values	Count of FactKey	Number of match records per team per year
Values	Sum of Sixes	Total sixes hit per team per year
Values	Sum of Balls	Total balls faced per team per year

Formatting Applied

- Row subtotals: Enabled — total per team across all years
- Column subtotals: Enabled — total per year across all teams
- Grand totals: Enabled — overall total row and column
- Column headers: Dark blue background, white text
- Row headers: Dark blue background, white text
- Visual title: *'Cricket Performance by Team and Year'* — centered, bold
- Page title text box: *'Matrix Visual: Cricket Performance Analysis'* — dark blue background, white text



4.3 Report 2: Cascading Slicers with Multiple Visuals

Overview

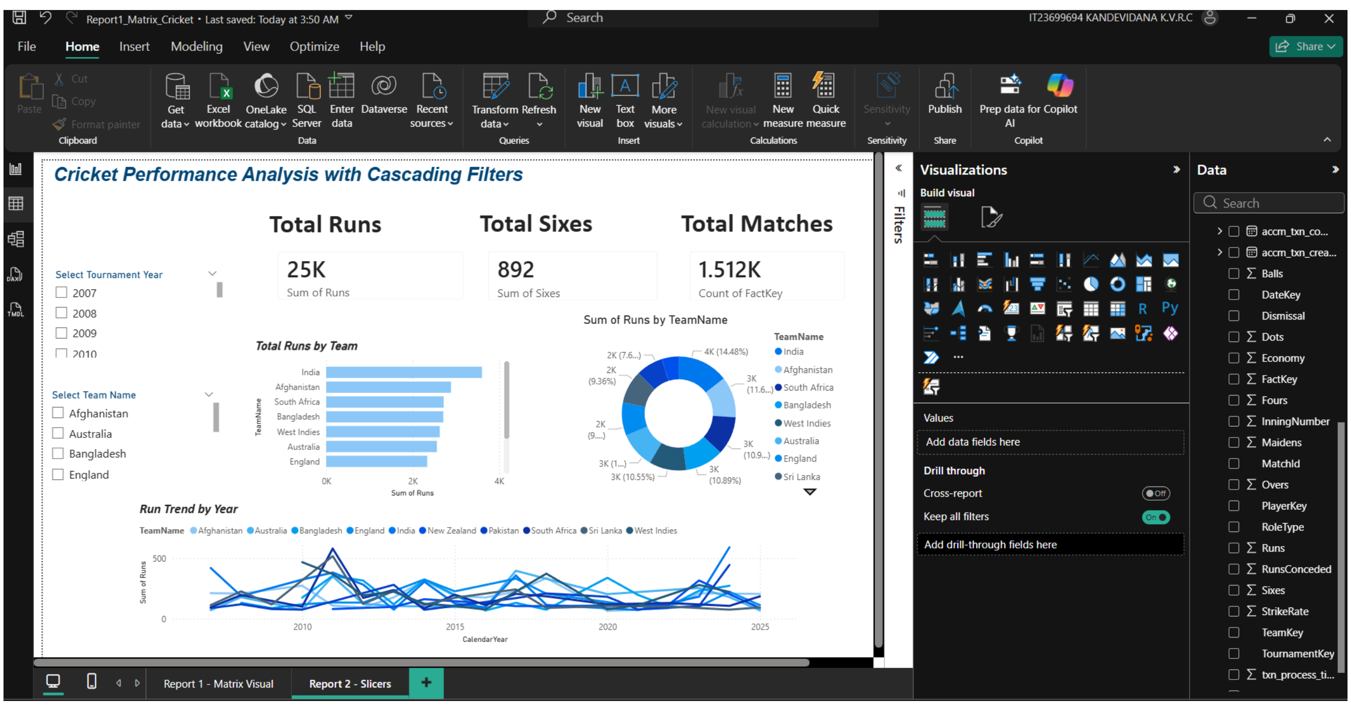
Report 2 implements cascading filters: selecting a value in Slicer 1 (Tournament Year) dynamically filters Slicer 2 (Team Name). Multiple charts provide comprehensive insights from the filtered cricket performance data.

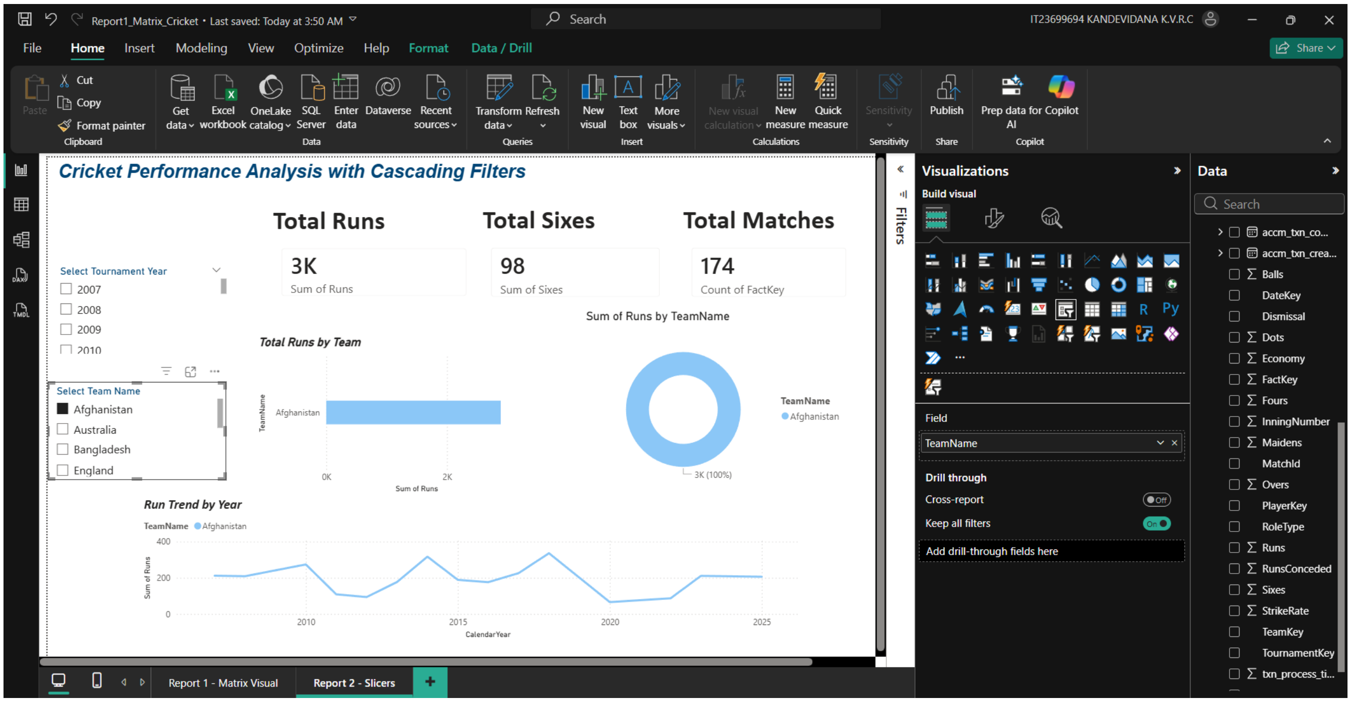
Cascading Filter Logic

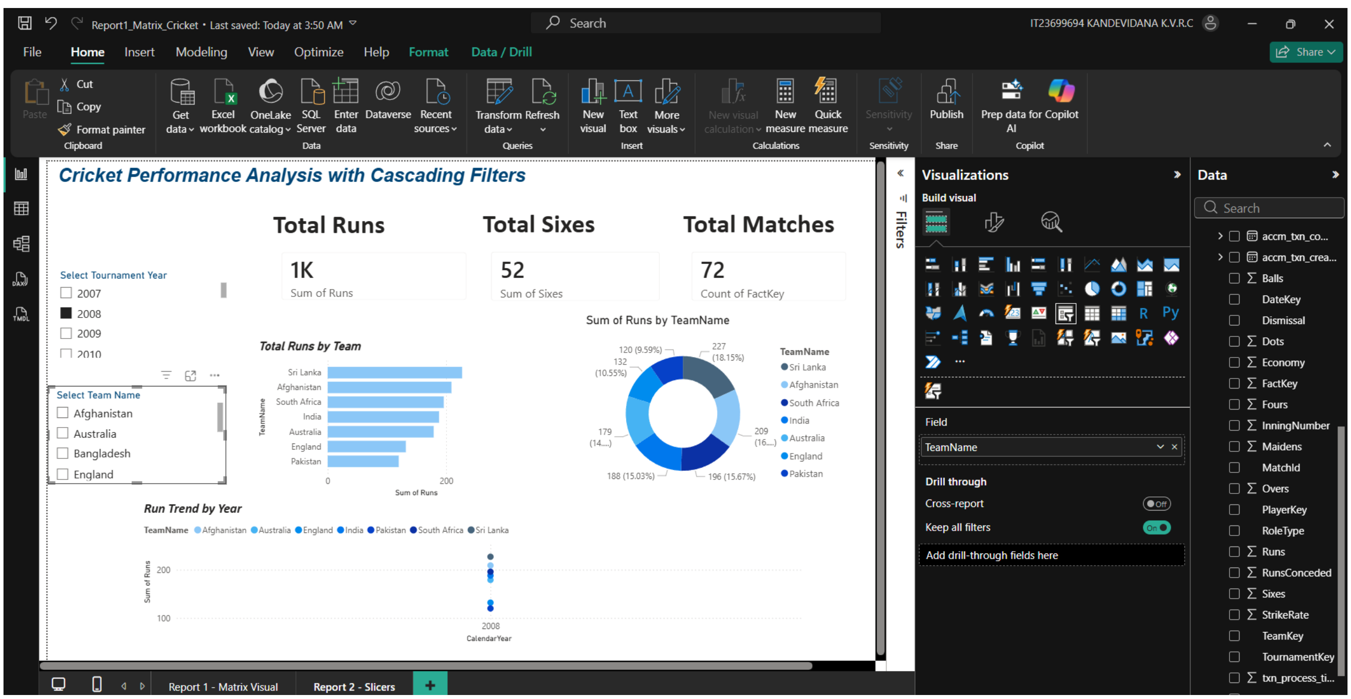
The cascading filter works because both slicers use fields from related dimensions — DimTournament and DimTeam — both connected to Match_Fact. When a specific Tournament Year is selected in Slicer 1, Slicer 2 automatically updates to show only the teams that participated in that tournament. For example, selecting 2007 in Slicer 1 filters Slicer 2 to show only teams that played in 2007 — all other teams are hidden.

Visuals on Report 2

Visual	Type	Fields	Purpose
Slicer 1	Slicer	Dim Tournament.Tournament Year	Filter by tournament year — drives Slicer 2
Slicer 2	Slicer	Dim Team.Team Name	Filter by individual team — cascades from Slicer 1
Card 1	Card	Runs (Sum)	KPI — Total Runs
Card 2	Card	Sixes (Sum)	KPI — Total Sixes
Card 3	Card	FactKey (Count)	KPI — Total Match Records
Bar Chart	Clustered Bar	TeamName vs Sum of Runs	Compare runs across all teams
Line Chart	Line Chart	CalendarYear vs Sum of Runs	Run trend across years by team
Donut Chart	Donut	TeamName vs Sum of Runs	Run distribution percentage by team







4.4 Report 3: Drill-Down Hierarchical Report

Overview

Report 3 allows hierarchical cricket data exploration using drill-down. Users navigate Year → Quarter → Month → Day by clicking the drill-down arrows on the column chart. The Date Calendar hierarchy from the SSAS cube drives the drill levels.

Drill-Down Controls

- Single down arrow (↓): Enable drill mode — click any bar to drill into next level

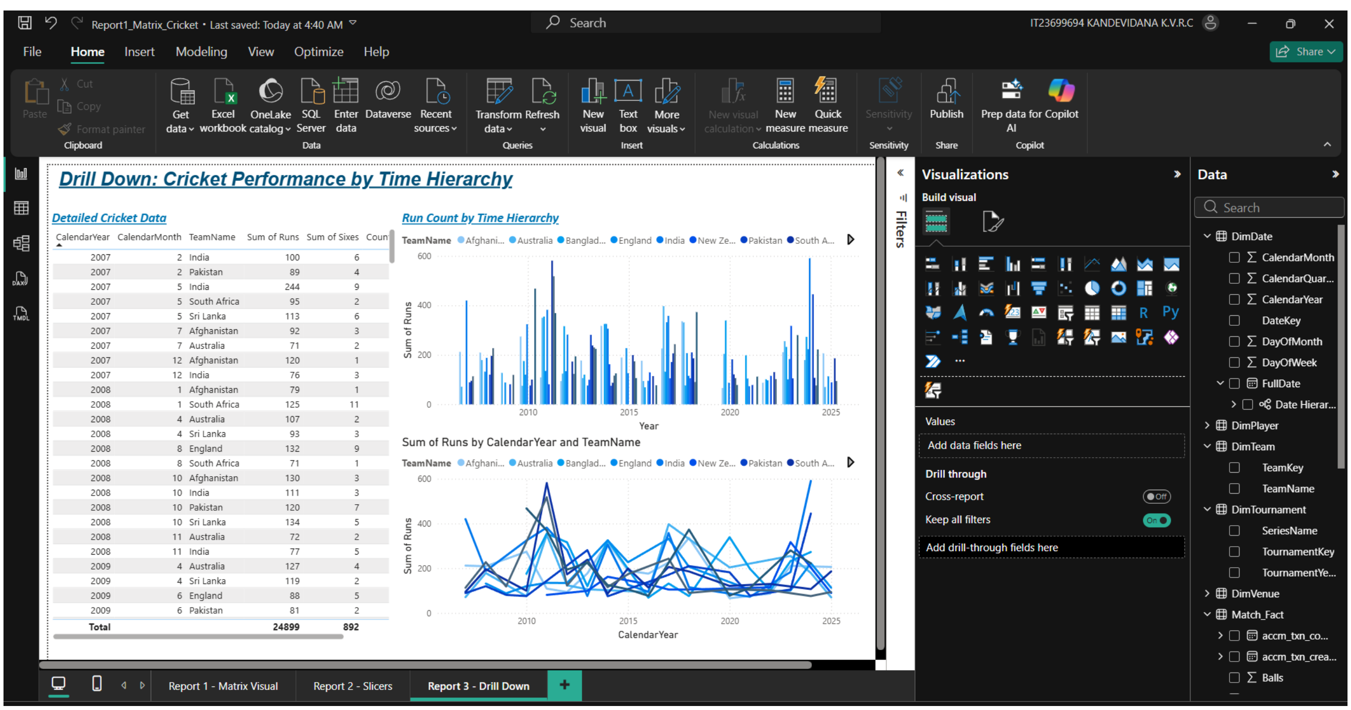
- Single up arrow (↑): Drill up to previous level
- Double down arrow (↓↓): Expand all — shows all hierarchy levels simultaneously
- Hierarchy path: Year (2007–2029) → Quarter (Q1–Q4) → Month (1–12) → Day

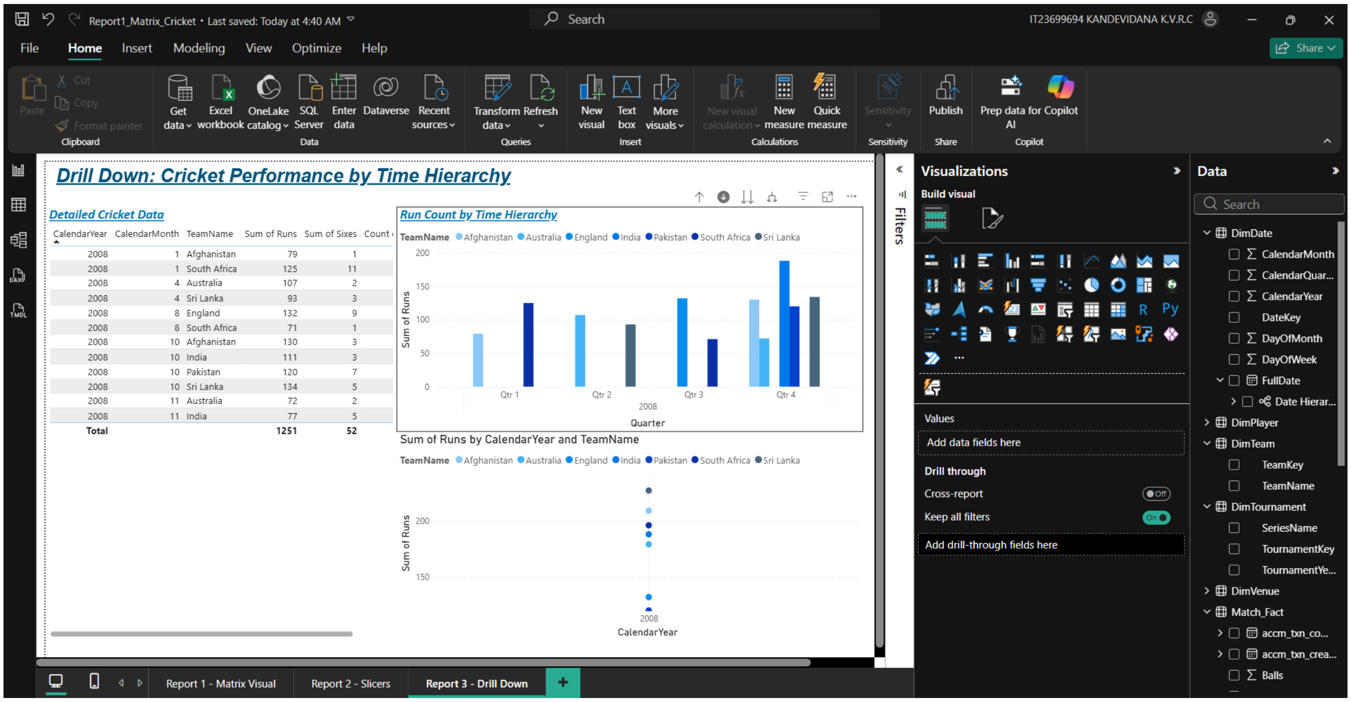
Visuals on Report 3

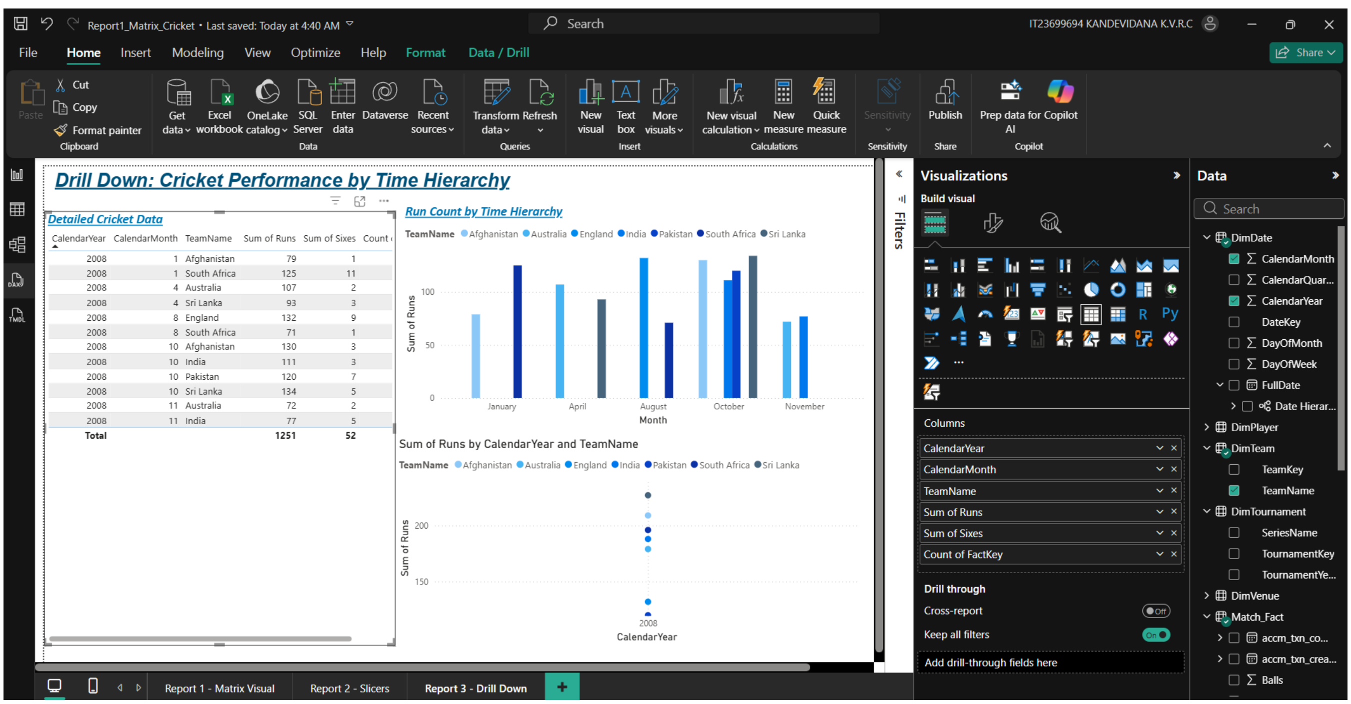
Visual	Fields	Purpose
Table Visual	Calendar Year, Month, Player Name, Team Name, Runs, Wickets, Match Fact Count	Detailed raw cricket data with all hierarchy levels visible — Total: 24,899 Runs, 892 Sixes
Column Chart (main)	X: CalendarYear/Quarter/Month/Day, Y: Sum of Runs, Legend: TeamName	Primary drill-down visual — click arrows to navigate hierarchy from Year down to Day level
Line Chart	X: CalendarYear, Y: Sum of Runs, Legend: TeamName	Trend analysis showing run performance across years (2007–2025) by team

Drill-Down Demonstration

- Level 1 — Year view: Shows total runs per year (2007–2029) grouped by team
- Level 2 — Quarter view: Expands to Q1, Q2, Q3, Q4 within each year
- Level 3 — Month view: Expands to individual months within each quarter
- Level 4 — Day view: Most granular level showing individual day-level data







4.5 Report 4: Drill-Through Report

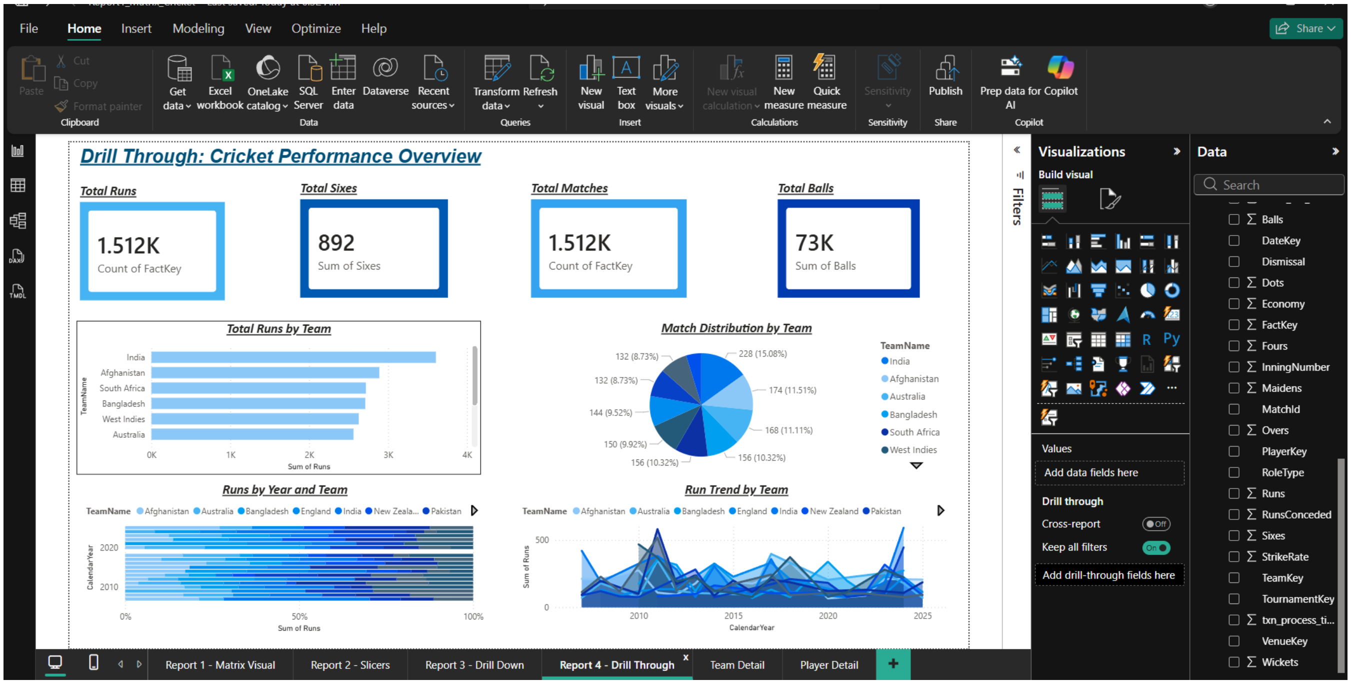
Overview

Report 4 implements drill-through navigation. Users right-click on any data point in the main overview page and navigate to a detail page showing data filtered to their selection. One drill-through destination page was created..

Report 4 Page Structure

Page	Name	Purpose
Page 1	Report 4 - Drill Through	Main overview — users right-click here to navigate
Page 2	Team Detail	Drill-through destination — detailed data for selected team
Page 3	Player Detail	Drill-through destination — detailed data for selected player

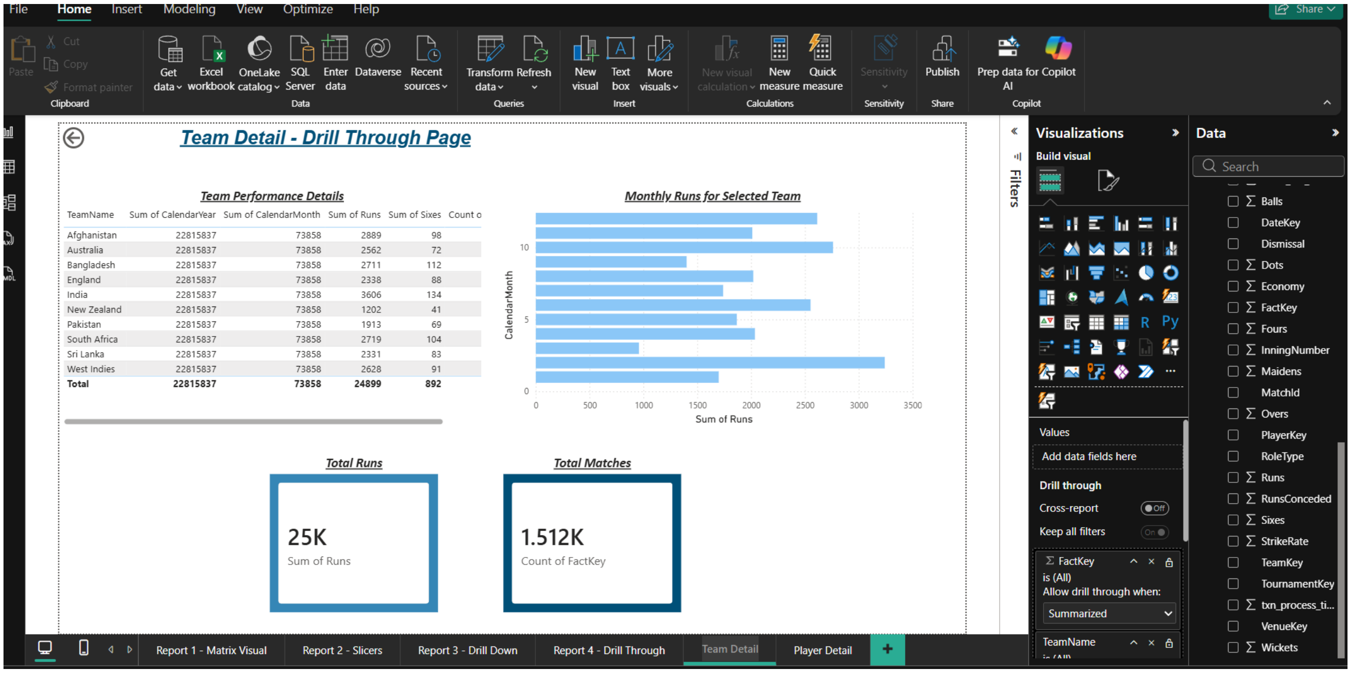
Page 1 — Main Overview Visuals



Page 2 — Team Detail

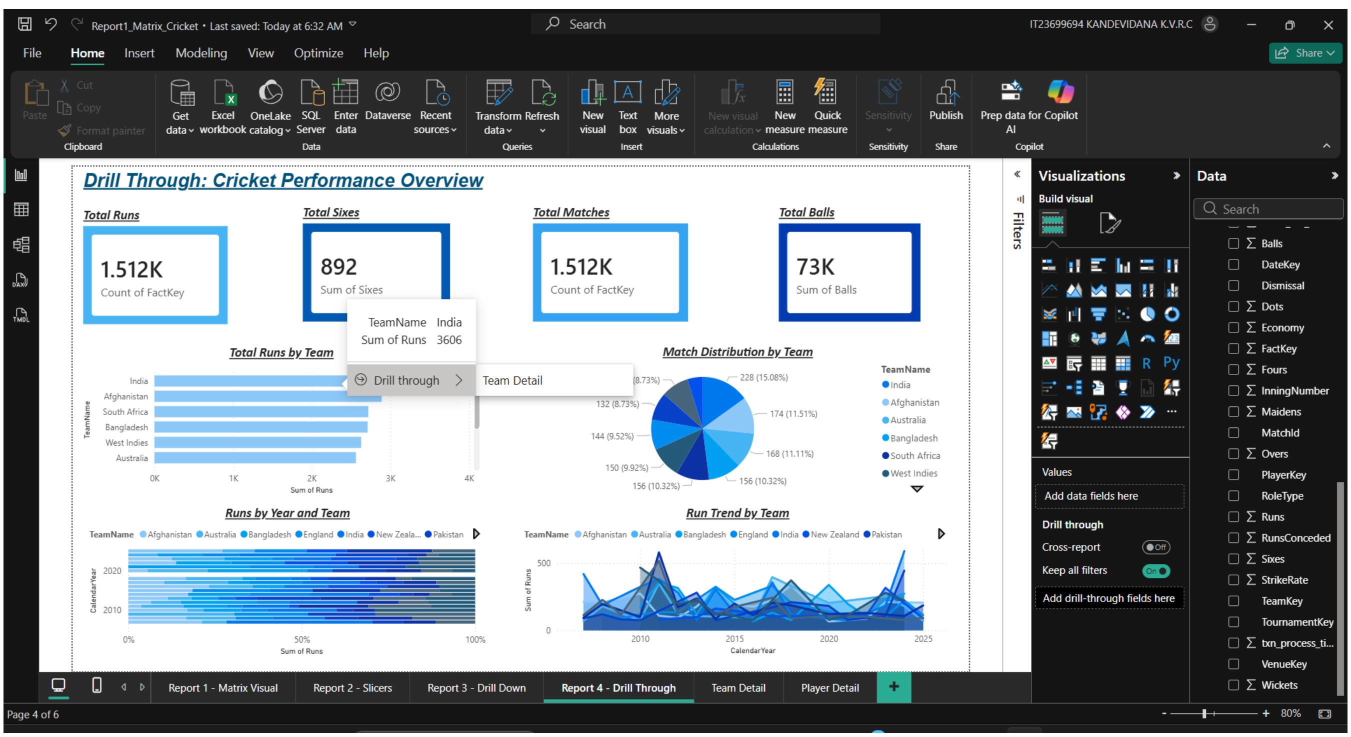
The Team Detail page was configured with TeamName in the Drill-Through field area. When users right-click a team on the main page and select Drill through → Team Detail, this page opens filtered to that team's data only.

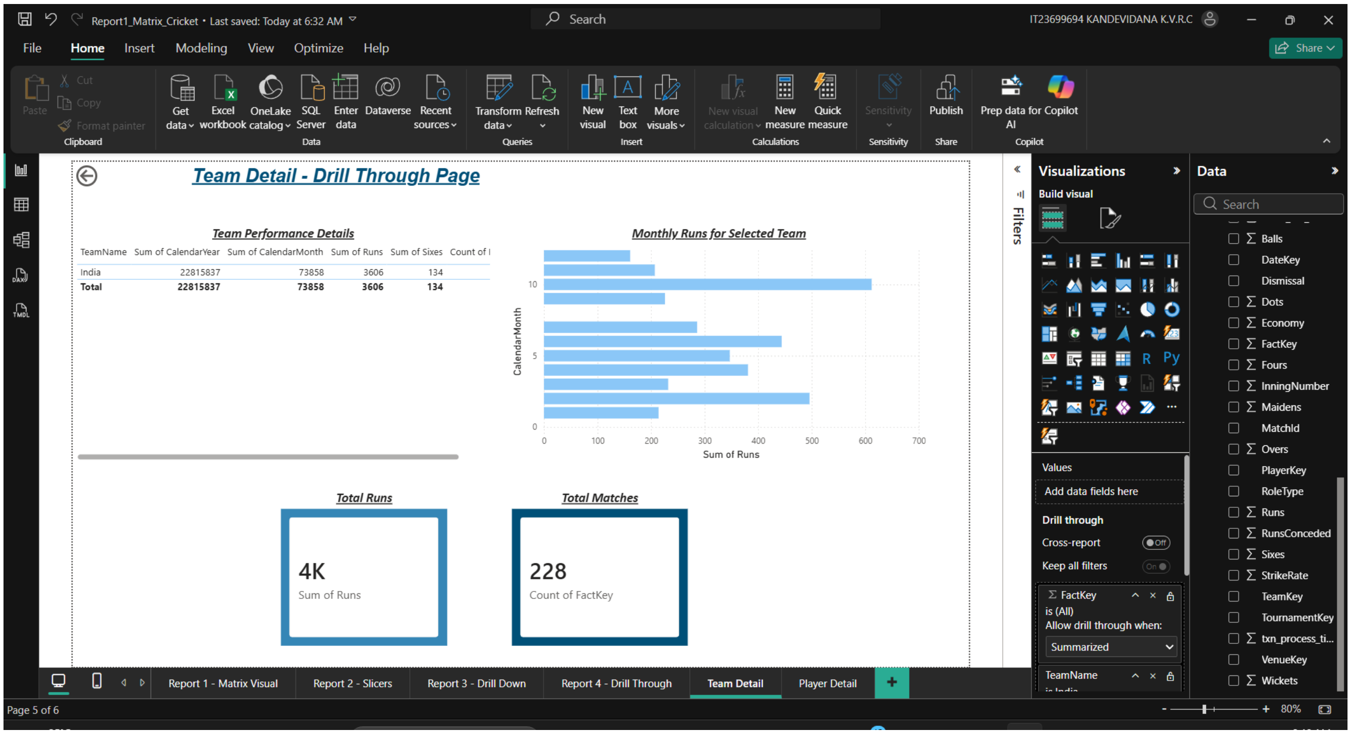
- Drill-through field: DimTeam.TeamName
- Visuals: Table (TeamName, CalendarYear, CalendarMonth, Runs, Sixes, Count of FactKey), Bar Chart (Monthly Runs), 2 Card visuals
- Back navigation arrow automatically appears at top left



Testing Drill-Through Navigation

Drill-through was tested by right-clicking on bar chart bars in the main page. The context menu appeared showing both Line Detail and Station Detail as drill-through options.





4.6 Power BI Service Publication

All four Power BI reports were published to Power BI Service using the Publish button. The file PowerBI_IT23699694.pbix was saved and published to My Workspace.

Publication Steps

1. Saved file as PowerBI_IT23699694.pbix
2. Clicked Home → Publish in Power BI Desktop ribbon
3. Selected 'My Workspace' as the destination workspace
4. Waited for publication to complete successfully
5. Clicked 'Open in Power BI' link to verify in browser
6. All 6 pages confirmed visible in Power BI Service: Report 1, Report 2, Report 3, Report 4, Player Detail, Team Detail
7.

Submission Folder Structure

Folder	Contents
DataWarehouse_IT23699694/	CricketDWBI_DW database backup (.bak file)
CubeProject_IT23699694/	CricketDWBI_Cube Visual Studio solution files (.cube, .dsv, .ds, .dim files)
Excel_IT23699694/	OLAP_Operations_IT23699694.xlsx — all 5 OLAP operations demonstrated
PowerBIReports_IT23699694/	PowerBI_IT23699694.pbix — all 4 reports (6 pages total)
Document_IT23699694/	Assignment2_IT23699694.pdf (this document) and .docx source
DWBI_Assignment_2_Answer_IT23699694/	Root folder containing all above subfolders — zipped as IT23699694.zip

End of Assignment 2